Muskan Kumar

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SUMMARY

Final-year B.Tech Computer Science and Engineering (AI & ML) student with hands-on project experience in AI-powered applications, multi-agent systems, and machine learning pipelines. Proficient in Python, Scikit-learn, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), and FastAPI, with practical experience in LLM integration, vector search, and ML model deployment.

EDUCATION

Shobhit Institute of Engineering & Technology | Meerut, Uttar Pradesh **2023–2027**
B.Tech in Computer Science and Engineering (AI & ML**)**
CGPA: 8.86/10

Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Database Management Systems, Operating Systems, Software Engineering

PROJECTS

AgentForge AI — Multi-Agent AI Workspace | GitHub **Sept 2026**
Python · FastAPI · LangGraph · LangChain · PostgreSQL · Pinecone

- Designed a multi-agent architecture using LangGraph for coding, research, conversation, and document processing.
- Implemented RAG, embeddings, and vector search for context-aware PDF question answering.
- Integrated web search, content generation, and persistent conversational memory.
- Secured FastAPI endpoints with authentication, input validation, rate limiting, and safety guardrails.

Enterprise HR Policy Copilot — Agentic RAG System | GitHub **Sept 2026**
Python · FastAPI · LangGraph · Pinecone · Sentence Transformers · Docker

- Built an agentic RAG pipeline to retrieve relevant HR policy information and generate document-grounded answers.
- Implemented evidence grading, conditional web search, and source citations to support answer verification.
- Added document ingestion, workflow tracing, and audit logging, containerized the application with Docker.
- Validated document-grounded responses, retrieval relevance, and source citations using policy-related test queries.

NIDS — Network Intrusion Detection MLOps Pipeline | GitHub **Oct 2026**
Python · Scikit-learn · MLflow · FastAPI · Docker

- Developed an end-to-end ML pipeline for network intrusion detection, including data ingestion, preprocessing, and model training.
- Used MLflow to track experiments and compare model performance.
- Built a FastAPI inference service and integrated data-drift detection into the pipeline.

TravelMate AI — Multi-Agent Travel Planning Assistant | GitHub **Aug 2026 - Sept 2026**
Python · LangGraph · LangChain · FastAPI · Groq · Tavily · PostgreSQL

- Developed a LangGraph-based multi-agent travel planner to process natural-language trip requirements and coordinate planning workflows.
- Integrated travel research tools for itinerary planning, flight information, weather, and budget estimation.
- Built a FastAPI backend to manage travel requests and orchestrate agent workflows.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java
- **Machine Learning & Data Science:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras
- **Generative AI & LLMs:** LangChain, LangGraph, RAG, LLM Integration, Sentence Transformers
- **Frameworks & APIs:** FastAPI, Flask, React.js, Node.js, Express.js, REST APIs
- **Databases & Vector Search:** MongoDB, MySQL, PostgreSQL, Pinecone
- **MLOps & Developer Tools:** MLflow, Docker, Git, GitHub, Postman, VS Code, Jupyter Notebook
- **Cloud & Deployment:** AWS S3, Render, Vercel, Netlify

CERTIFICATIONS

- **Intro to Machine Learning** — Kaggle
- **AI Agents Course** — Hugging Face
- **Introduction to Generative AI** — Google Cloud
- **Introduction to Large Language Models** — Google Cloud
- **Generative AI Fundamentals** — Google Cloud